

# Choose Informative Questions

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can fit a root question and leaf predictions from training examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

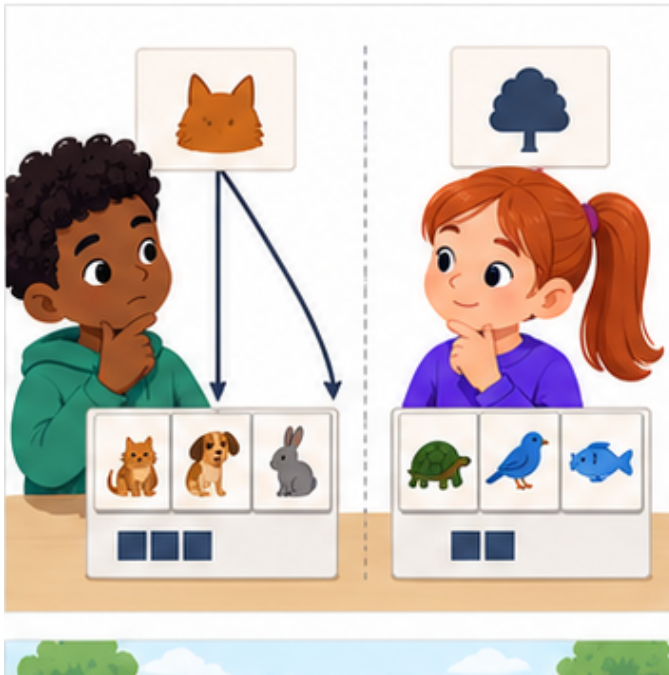
Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 15: Choose Informative Questions

A strong question eliminates many possibilities without relying on accidents.



## THINKER MISSION

Compare two first questions by remaining possibilities.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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## Readiness for the connected explanation

Why does a cutoff with zero training errors still need validation on separate examples?

## This week's end goal

By the end, I can choose a question that separates possibilities rather than repeating a constant clue.

# 2. Notice the evidence

## PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use choose informative questions to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

**Write or draw your reasoning and evidence.**

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My prediction and the clue supporting it:

**Write or draw your reasoning and evidence.**

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## A closer explanation

A decision tree asks a feature question and follows a branch to a prediction. A tree with one question has depth one and is sometimes called a stump. Unlike Week 8, where the two output labels were fixed in advance, this week we fit both the question and the label stored at each leaf.

Our six fictional training rows have two binary sensor flags available before prediction: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate. Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

### Row · Near · Tall · Label

**Row: 1**

**Near: 1**

**Tall: 0**

**Label: blocked**

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**Row: 2**

**Near: 1**

**Tall: 1**

**Label: blocked**

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**Row: 3**

**Near: 1**

**Tall: 0**

**Label: blocked**

---

**Row: 4**

**Near: 0**

**Tall: 1**

**Label: clear**

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**Row: 5**

**Near:** 0

**Tall:** 0

**Label:** clear

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**Row:** 6

**Near:** 0

**Tall:** 1

**Label:** blocked

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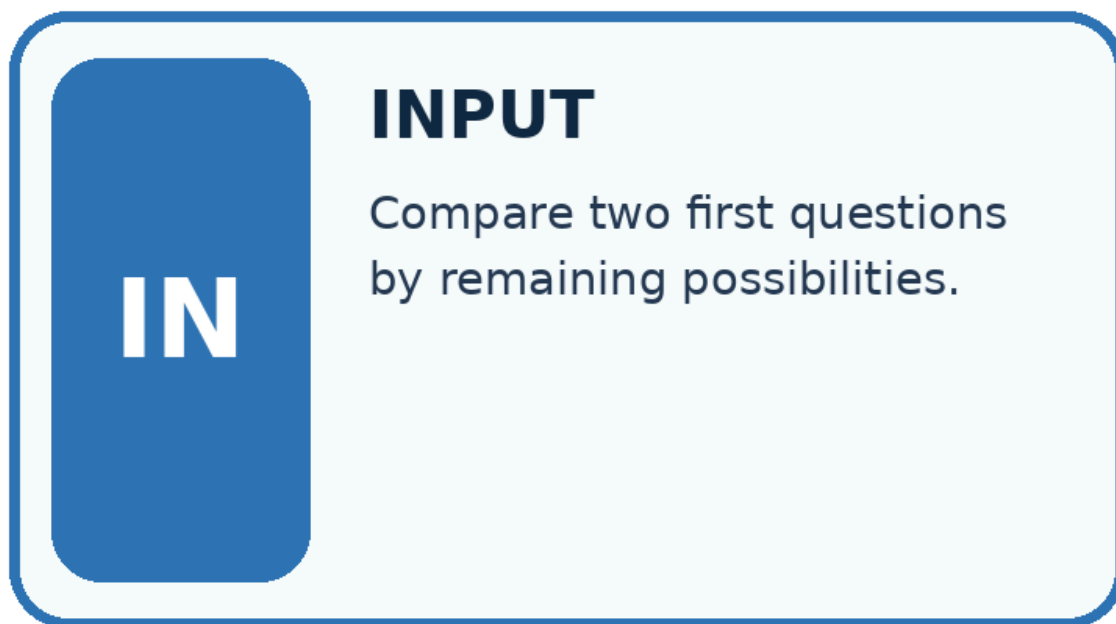
Try near first and tall second. For each question, group the training rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts. Count all training mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom fitting procedure.

### 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

A strong question eliminates many possibilities without relying on accidents.



OUT

## OUTPUT

Compare two first questions by remaining possibilities.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]

def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"

best_errors = len(rows) + 1

for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)

    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1

    print(feature, errors)

    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)

print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. Output is 0 1, then 1 2, then 0 clear blocked. Near makes one training mistake: near=0 contains two clear labels and one blocked, so that leaf predicts clear. Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both branches of each candidate. An empty branch would need a declared fallback; this small example does not implement one. The score is total training mistakes, not an average giving small and large branches equal weight.

This is an explicitly limited depth-one learner. General tree algorithms can repeat split selection within branches and often use impurity criteria such as Gini. Our code does not implement that full procedure. Keeping the scope small lets you inspect every decision.

### Connect the picture and the explanation

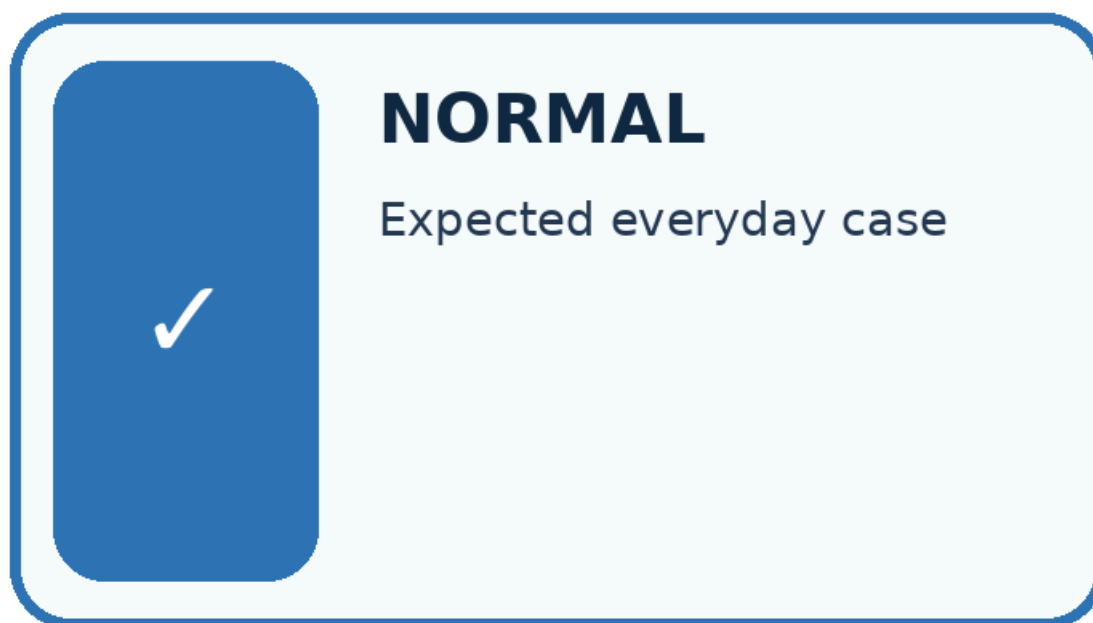
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

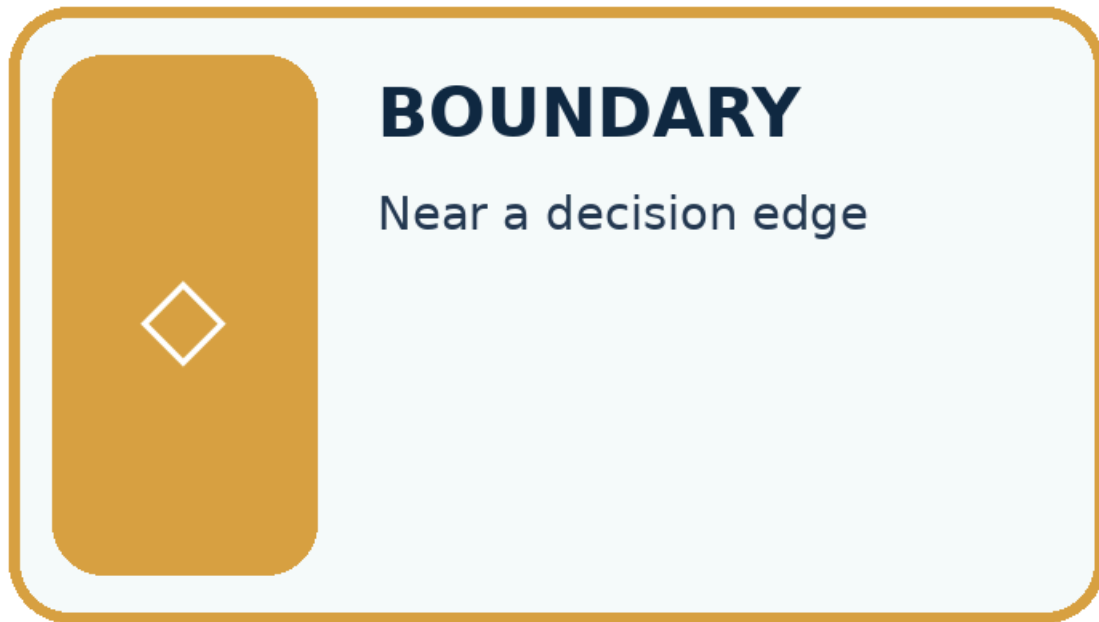
## 4. Compare the cases

### CASE GALLERY

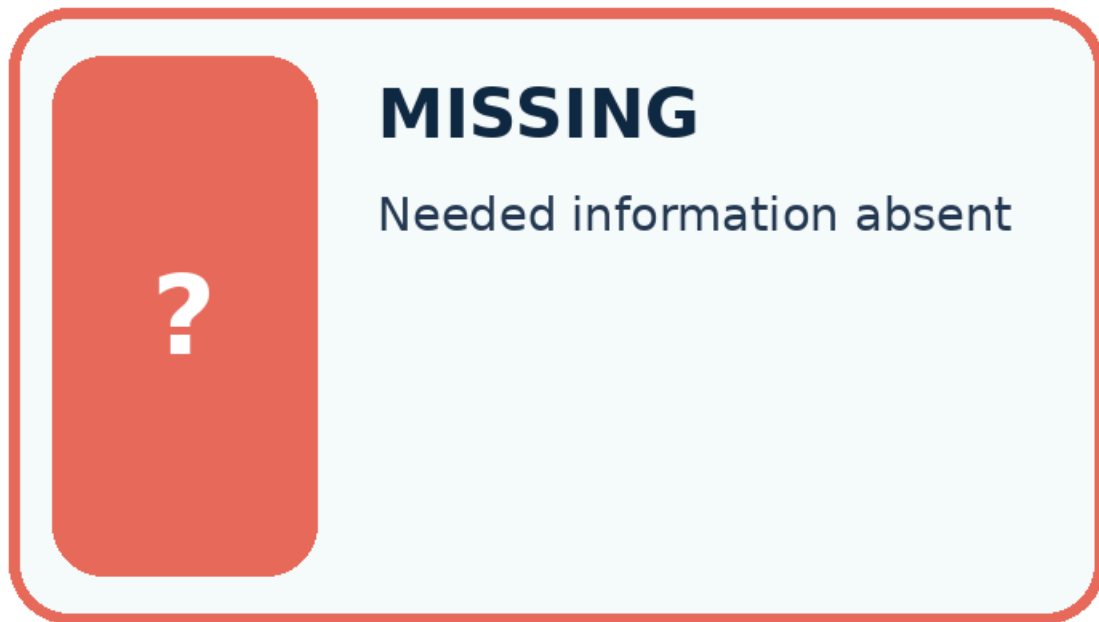
#### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.





MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: The current possibility set contains only the four mystery cards.

Prediction: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

**1**

## **INFORMATIVE**

reduces uncertainty usefully

**2**

## **SPLIT**

groups produced by a question

3

## REMAINING POSSIBLE

candidates still possible

4

## EFFICIENCY

useful result with fewer steps

**Word · My own example · Not this / boundary**

**Word:** informative

**My own example:** Providing evidence that helps distinguish relevant possibilities or reduce uncertainty.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** split

**My own example:** Separate a dataset into portions with different development or evaluation roles.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** remaining possibilities

**My own example:** The options still consistent with the evidence after some have been ruled out.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** efficiency

**My own example:** The resources required to achieve a result, such as time, memory or number of steps.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Root

**Meaning:** The first question asked by a tree

\_\_\_\_\_

**Word:** Branch

**Meaning:** A path selected by the answer to a question

\_\_\_\_\_

**Word:** Leaf

**Meaning:** An endpoint that stores a prediction

\_\_\_\_\_

**Word:** Depth

**Meaning:** The largest number of question steps from root to leaf

\_\_\_\_\_

## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

## WEEK 15 • WORKED EVIDENCE

- 1 Eight cards remain.
- 2 'Has wings?' splits 4/4; worst branch leaves 4.
- 3 'Is red?' splits 1/7; worst branch leaves 7.
- 4 For this set, wings is the more informative first question.

1. Eight cards remain.
2. 'Has wings?' splits 4/4; worst branch leaves 4.
3. 'Is red?' splits 1/7; worst branch leaves 7.
4. For this set, wings is the more informative first question.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

#### Two paths through the learned tree

**Root question · Branch answer · Learned leaf**

**Root question:** Is near equal to 1

**Branch answer:** No near is 0

**Learned leaf:** Predict clear

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**Root question:** Is near equal to 1

**Branch answer:** Yes near is 1

**Learned leaf:** Predict blocked

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Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: predictions are blocked, clear, clear, blocked. Two of four match. The training-majority baseline predicts blocked, since training has four blocked and two clear; it also matches two of four validation labels.

**Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.**

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Keep the fitted root and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct training predictions out of six is not the validation result.

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Choose Informative Questions

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

## The facts we will use

Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

## Worst-case remaining candidates

cards remaining



Bar length shows the supplied value; the marked baseline is zero.

## Choose Informative Questions

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. List possibilities

There are eight possible target cards initially.



#### 2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



#### 3. Paperback question

Yes 6; No 2; largest remaining group 6.



#### 4. English question

Yes 8; No 0; largest remaining group 8.

### CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. List possibilities

**What happens and why:** There are eight possible target cards initially.

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**Step:** 2. Mystery question

**What happens and why:** Yes group 4; No group 4; largest remaining group 4.

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**Step: 3.** Paperback question

**What happens and why:** Yes 6; No 2; largest remaining group 6.

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**Step: 4.** English question

**What happens and why:** Yes 8; No 0; largest remaining group 8.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set. This is an identification-game criterion; a classifier fitting labels may instead compare prediction errors or another declared training measure.

## What this example does not establish

Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Read the labelled picture: Choose the best original question and explain why its more balanced split matters here.

Change one thing: The current possibility set contains only the four mystery cards. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: The most unusual question is always the most informative. Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Record the two candidate questions, branch label counts, total errors, tie rules, depth limit, stored tree and validation predictions. Retain all rows, including the blocked example in the near=0 branch; removing a disagreement would change the evidence.

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset. Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

**A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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### A closer explanation

Do not assume near=1 must always predict blocked. If a new training set gives that branch a clear majority, this same learner will store clear there. A feature name is not the answer. Use the training labels belonging to that branch.

Adding more questions can fit finer training details but does not guarantee better new-case performance. Record any depth choice made during development, and keep final test data separate. A leaf majority describes this training group; it is not a promise about every future example.

### Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example “Fitting a decision tree”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Choose the best original question and explain why its more balanced split matters here.

**My explanation using a pictured step and the given facts**

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### **PREDICT → TEST → RECORD → EXPLAIN**

For each proposed question, count yes/no groups and the largest remaining group. Choose, justify, then test on a changed card set.

For each proposed question, count yes/no groups and the largest remaining group. Choose, justify, then test on a changed card set.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

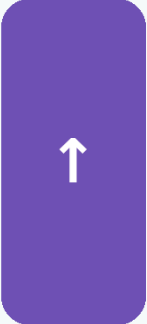
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



**MASTERY LADDER**  
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Ask first about preferences that remove the largest number of unsuitable books.

THIS WEEK'S CHECKPOINT Ask first about preferences that remove the largest number of unsuitable books.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



**INDEPENDENT PROOF**  
Compare two first questions by remaining possibilities.

CHALLENGE Compare two first questions by remaining possibilities.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Eight blue cards contain four circles and four triangles; target is shape.

#### The question

Which question is more useful: blue or circle?

#### Worked explanation

Circle separates target groups. Blue puts every card in one branch. Usefulness depends on the task and examples.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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### Trace the supplied case visually

### Choose a question that separates the target

**Evidence or stage · Value or result**

**Evidence or stage:** Blue?

**Value or result:** All eight stay together

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**Evidence or stage:** Circle?

**Value or result:** Four circles versus four triangles

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**Evidence or stage:** Target

**Value or result:** Shape, not colour

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Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Circle separates target groups. Blue puts every card in one branch. Usefulness depends on the task and examples.

## 14. Try a new case independently

### Independent case: Changed independent case

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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## Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Ask first about preferences that remove the largest number of unsuitable books.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

## By the end, I can · How I will show it

**By the end, I can:** Fit a root question and leaf predictions from training examples

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Trace a new example through the learned tree

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Compare a shallow tree with a baseline and state its limits

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Fit a root question and leaf predictions from training examples

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.